

RRGE-RAG:Radiology Report Generation Enhancement Using Hybrid Retrieval Augmented Generation

Victor Michel Benadict Supipi Sudeshika VirajiniKarunathilaka

Vibolrottana Seng

Dakchhyeta Bade Shrestha

Zwe Yu Ya Kyaw Zin Oo

Asian Institute of Technology, Faculty of Advanced Science and Technology (FAST)

{st126489,st126425,st126671,st125990}@ait.ac.th

Abstract

Automated radiology report generation is an important application of medical artificial intelligence that aims to reduce radiologists' workload and improve diagnostic efficiency. This final report presents a Hybrid Retrieval-Augmented Generation framework for chest X-ray report generation that integrates visual features, structured clinical knowledge, and retrieved case-based evidence to improve the factual correctness and coherence of generated reports. The proposed approach extends the R2Gen architecture by incorporating BioViL-T as the image encoder and applying feature alignment with attention-based fusion to combine visual embeddings, retrieved reports, and clinical labels. Experiments are conducted on the Indiana University Chest X-ray dataset and compared with baseline models implemented using their original source code for fair evaluation. The system is evaluated using lexical metrics, including BLEU and ROUGE-L; clinical metrics, including CheXbert F1 and RadGraph F1; and uncertainty-aware metrics, including Expected Calibration Error and Brier Score. In addition, this report analyzes how uncertainty estimates relate to generation quality and model calibration. The results provide insights into the strengths and limitations of retrieval-augmented radiology report generation and its potential for developing more reliable and clinically applicable medical AI systems.

1 Introduction

A major medical application of natural language generation (NLG) is the development of assistive systems that analyze patient X-ray images and automatically produce a textual report describing the clinical findings visible in the images. (Jing et al., 2018) However, medical report writing requires specialized domain expertise, and only experienced radiologists can accurately interpret chest X-ray images and describe the corresponding

findings in a coherent manner. ¹An automatic chest X-ray report generation system can reduce the workload of radiologists and are in urgent need (Brady et al., 2012). Recent multimodal foundation models have exhibited remarkable capabilities in challenging healthcare tasks, motivating an automation of this process to enhance physicians' efficiency on clinical decision-making and improve patient health outcomes. (Sun et al., 2025) newline Although prior medical multimodal foundation models have demonstrated promising capabilities on report generation given the radiology image, they still suffer from serious hallucinations by generating factually inaccurate reports (Pal et al., 2023; Pal and Sankarasubbu, 2024; Ahmad et al., 2023). Factual correctness is especially critical in chest radiology domains, as minute textual differences can drastically invert radiology report meaning and downstream prescribed treatments (Delbrouck et al., 2022; Song et al., 2024; Liu et al., 2024). Retrieval-Augmented Generation (RAG) has emerged as a popular paradigm to address this issue by grounding text generation with retrieved relevant knowledge given a query (Lewis et al., 2020; Wolf et al., 2024; Gao et al., 2023). However, developing medical multimodal retrievers remains challenging, requiring retrievers to bridge the gap between symptomatic image semantics and factually-equivalent report text.

To capture fine-grained details in chest radiographs and improve the factual completeness of generated reports we introduce **RRGE-RAG** it is framework takes as input a patient's chest X-ray images, including both frontal and lateral views, together with structured clinical labels derived from the Problems column. Given these

¹<https://github.com/Vibolrottanaseng/NLU-Project-Wave2Vec>

inputs, the system performs retrieval to identify the Top-3 most similar prior patient cases based on image similarity. Each retrieved case provides complementary contextual information, including the corresponding frontal and lateral chest X-ray images, the associated radiology report composed of Findings and Impression sections, and the structured clinical labels. By leveraging these retrieved examples as external evidence, the model is guided to generate a clinically grounded report for the input case. The primary generation target is the Impression section, since it summarizes the key diagnostic conclusion in a concise and clinically useful form, while generation of the Findings section can be considered as an optional extension. This design allows the framework to combine visual evidence from the current patient with relevant knowledge from similar historical cases, with the goal of improving factual accuracy, reducing hallucination, and producing more reliable radiology reports as shown in Figure 1.

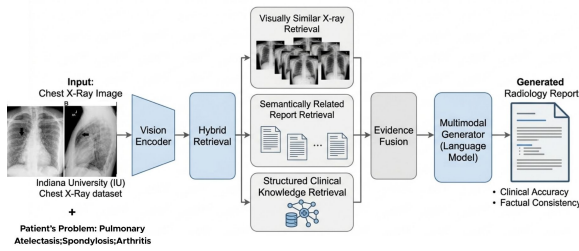


Figure 1: An Overview of RRGE-RAG framework. Hybrid RAG framework for chest X-ray report generation. A chest X-ray from the Indiana University dataset is passed through a vision encoder, then a hybrid retrieval module retrieves three types of supporting evidence: visually similar X-rays, semantically related reports, and structured clinical knowledge. These retrieved evidences are combined through evidence fusion and sent to a multimodal language generator, which produces the final radiology report with improved clinical accuracy and factual consistency.

We have came up with 3 research question for this study:

1. How does the integration of retrieval-augmented generation (RAG) influence the accuracy and clinical relevance of automated radiology report generation compared to traditional image-to-text models? This question evaluates whether adding retrieval improves report generation beyond standard image-to-

text models such as R2Gen. Traditional models rely only on learned image-text mappings, which can lead to incomplete or hallucinated findings. By retrieving similar prior cases, the proposed method aims to ground generation in relevant external evidence. This question can be answered by comparing the proposed model with baseline methods using BLEU and ROUGE-L, while also examining whether retrieval leads to more clinically relevant and factually consistent reports.

2. To what extent does multimodal fusion of X-ray images, clinical labels, and retrieved case information improve the coherence and completeness of generated radiology reports? This question focuses on the effect of combining multiple information sources in the proposed framework. The model integrates image features, structured clinical labels, and retrieved case information to provide richer context for report generation. This can be examined through ablation studies that compare image-only input with versions that add labels, retrieval, and full multimodal fusion. Better performance from the full model would indicate that combining these modalities improves report coherence and completeness.
3. How well do uncertainty estimates (measured using entropy and MC dropout variance) correlate with generation quality (BLEU, ROUGE), and how well are they calibrated (ECE, Brier Score)? This question examines whether the model’s uncertainty estimates are meaningful and reliable. If higher entropy or MC dropout variance is associated with lower BLEU or ROUGE scores, then uncertainty can serve as an indicator of poor-quality generation. Calibration metrics such as ECE and Brier Score further assess whether the model’s confidence matches its actual performance. This is important for clinical settings, where confidence information can help determine whether a generated report should be trusted or reviewed more carefully.

We expect that integrating retrieval mechanisms and multimodal fusion will significantly improve the quality of generated reports. Specifically, the proposed approach is anticipated to produce more accurate, coherent, and clinically relevant reports by leveraging contextual knowledge from similar

cases. Additionally, the uncertainty estimation module is expected to provide meaningful confidence scores that correlate with the correctness of the generated outputs. Overall, the system should outperform traditional approaches in both quantitative metrics (e.g., BLEU, ROUGE, ECE, Brier Score) and qualitative clinical assessment.

2 Experimental Settings

This section describes the overall methodology of the proposed system for automated radiology report generation. The pipeline follows a multimodal, retrieval-augmented framework that integrates image features, clinical knowledge, and retrieved case information. An overview of the system is illustrated in Figure 1, which presents the end-to-end flow from input data to report generation and evaluation.

2.1 Dataset

We use the **Indiana University Chest X-ray dataset**² for this study. This dataset consists of chest X-ray images and corresponding radiology reports for **3,851 patients**. For each patient, two views are typically available: a *frontal view* and a *lateral view*, which provide complementary anatomical information for clinical interpretation. In addition to images, the dataset includes a CSV file (`indiana_reports.csv`) containing structured and unstructured clinical information. The main columns used in this work are:

- **Problems:** A set of clinically relevant findings identified from the X-ray, representing medical conditions or abnormalities (e.g., cardiomegaly, effusion). These are derived from expert-written radiology reports.
- **Image:** Metadata describing the type or view of the X-ray (e.g., PA, lateral).
- **Indication:** The clinical reason for performing the X-ray (e.g., symptoms or suspected condition).
- **Comparison:** Information about prior studies used for comparison, if available.
- **Findings:** A detailed descriptive section of the radiology report explaining observations in the X-ray.

²<https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university>

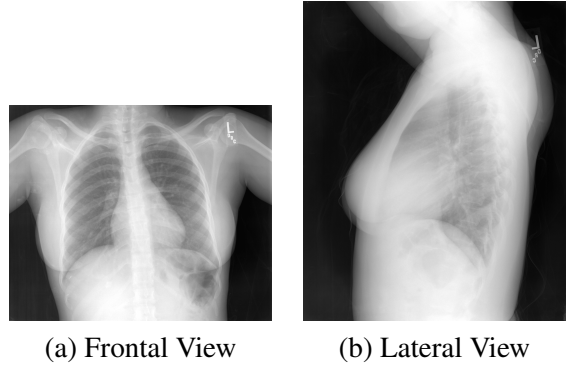


Figure 2: Example chest X-ray images from the IU dataset showing (a) frontal and (b) lateral views for a single patient.

- **Impression:** A concise summary of the most important clinical conclusions derived from the findings.

For our methodology, the inputs consist of **two X-ray images (frontal and lateral views)** along with the corresponding **Problems** (clinical labels), which provide structured medical knowledge. The target output is the **Findings** section, which serves as the ground truth radiology report to be generated by the model.

2.2 Data Preprocessing

Data preprocessing is a critical step to ensure consistency, data quality, and effective integration of multimodal inputs within the proposed **Hybrid Retrieval-Augmented Generation (Hybrid RAG)** framework. To maintain fairness across all experiments, the same preprocessing pipeline is applied to both baseline models and the proposed method.

Image Preprocessing For each patient, chest X-ray images are organized to include both *frontal* and *lateral* views, which provide complementary anatomical information. All images are resized and normalized according to the input requirements of the BioViL-T vision encoder. These standardized images are then used to extract domain-specific visual embeddings for both feature learning and retrieval.

Text Preprocessing For report generation, the *Findings* section is used as the target output. The text is preprocessed by converting to lowercase, removing unnecessary special characters, and normalizing whitespace while preserving clinically relevant terms. A word-level tokenizer is applied to

construct the vocabulary, and special tokens such as <start>, <end>, and <pad> are introduced. Each report is then converted into a sequence of token IDs and padded or truncated to a fixed maximum length. For text-based retrieval, reports are minimally preprocessed and encoded into dense semantic representations using a biomedical language model (e.g., BiomedBERT). This enables retrieval based on semantic similarity rather than exact lexical matching.

Structured Clinical Label Processing The *Problems* column is used to extract structured clinical knowledge. Each entry may contain multiple conditions, which are separated and converted into a multi-hot encoded vector representing the presence or absence of each condition. These structured labels are incorporated both as model inputs and as an additional signal in the hybrid retrieval process.

Data Filtering To ensure data quality and consistency, samples that do not meet the following criteria are excluded:

- (i) missing or empty *Problems* labels,
- (ii) absence of either frontal or lateral images, and
- (iii) missing or invalid *Findings* reports.

After filtering, the final dataset consists of 3,337 samples with aligned image, text, and structured label information.

Dataset Splitting and Consistency The dataset is divided into training (70%), validation (10%), and test (20%) sets at the patient level to prevent data leakage. Same train, validation, and test splits are used across all baseline models and the proposed Hybrid RAG framework. This ensures a fair and consistent evaluation by eliminating variability due to different data distributions. As a result, any observed performance differences can be attributed to the model architectures and methodological contributions rather than inconsistencies in data partitioning.

2.3 Baselines

To evaluate the effectiveness of the proposed **Hybrid Retrieval-Augmented Generation (Hybrid RAG)** framework, we compare it with several established baseline models:

- **R2Gen** (Chen et al., 2020): A standard image-to-text model for radiology report generation.

- **R2GenCMN** (Chen et al., 2021): An extension of R2Gen that incorporates cross-modal memory for improved generation.
- **PPKED** (Liu et al., 2021): A knowledge-enhanced model that integrates external medical knowledge.
- **METransformer** (Wang et al., 2023): A transformer-based architecture with multiple expert tokens.

These baselines represent diverse categories of approaches, including standard, memory-based, knowledge-enhanced, and modern transformer-based architectures, enabling a comprehensive and fair comparison with the proposed method.

Our proposed approach builds upon **R2Gen as the report generation backbone** and extends it with a **hybrid Retrieval-Augmented Generation (RAG) framework**. For image representation and retrieval, we employ **BioViL-T** (Bannur et al., 2023) as the image encoder to extract domain-specific visual embeddings from chest X-rays. For textual retrieval, we utilize **BiomedBERT (PubMedBERT-style) sentence embeddings** (Gu et al., 2021) to capture semantic similarity between radiology reports.

2.4 Evaluation Metrics

To ensure a comprehensive evaluation of the proposed framework, we employ three categories of metrics: lexical, clinical, and uncertainty-aware calibration metrics. These metrics jointly assess linguistic quality, clinical correctness, and prediction reliability.

Lexical Metrics We use standard text generation metrics to evaluate the similarity between generated and reference reports:

- **BLEU (BLEU-2, BLEU-3, BLEU-4)** (Papineni et al., 2002): Measures n-gram overlap between generated and ground truth reports at different granularities.
- **ROUGE-L** (Lin, 2004): Captures structural similarity by computing the longest common subsequence between generated and reference texts.

Clinical Metrics While lexical metrics evaluate surface-level similarity, they do not guarantee clinical correctness. Therefore, we employ clinically grounded evaluation metrics:

CheXbert F1 Score. (Smit et al., 2020) CheXbert (?) is used to convert generated reports into structured disease labels. We consider 14 clinically relevant chest conditions, commonly used in prior work (?):

- Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, Pneumothorax

From these labels, we compute four variants of F1 scores:

- **14Ma-F1 (14-class Macro F1):** Computes F1 independently for each of the 14 diseases and averages them equally. This treats all diseases equally, including rare conditions.
- **14Mi-F1 (14-class Micro F1):** Aggregates predictions across all 14 labels and computes a global F1 score. This is influenced more by frequent diseases.

The use of 14-label settings enables evaluation across both *comprehensive disease coverage* and *high-frequency clinical conditions*, ensuring balanced assessment of model performance.

RadGraph F1 Score (RG-F1). RadGraph (Jain et al., 2021) evaluates the correctness of extracted medical entities (e.g., observations and anatomical locations) and their relationships. Unlike CheXbert, which focuses on classification, RadGraph captures the structural and relational correctness of clinical information in generated reports.

Together, CheXbert and RadGraph provide a robust evaluation of **clinical accuracy**, ensuring that generated reports are not only fluent but also medically meaningful.

Uncertainty and Calibration Metrics In clinical settings, it is essential not only to produce accurate predictions but also to estimate their reliability. To evaluate model uncertainty, we employ the following metrics:

Expected Calibration Error (ECE). ECE (Posocco and Bonnefoy, 2021) measures how well predicted confidence scores align with actual correctness. Predictions are grouped into M confidence bins, and the difference

between average confidence and accuracy is computed:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|$$

where B_m is the set of samples in bin m , $acc(B_m)$ is the accuracy, and $conf(B_m)$ is the average confidence.

A lower ECE indicates better calibration, meaning the model’s confidence reflects true correctness.

Brier Score. (Redelmeier et al., 1991) The Brier Score measures the accuracy of probabilistic predictions by computing the mean squared error between predicted probabilities and ground truth labels:

$$Brier = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

In our setting, each generated report is converted into 14 binary disease labels using CheXbert. The model produces a probability p_i for each label, and the ground truth $y_i \in \{0, 1\}$. The Brier Score is computed across all labels and averaged over the dataset.

This formulation adapts naturally to the multi-label classification setting, allowing us to evaluate how well the predicted probabilities reflect true clinical outcomes.

Interpretation for Uncertainty Evaluation ECE and Brier Score provide complementary insights into model uncertainty:

- Low ECE indicates that confidence estimates are well calibrated.
- Low Brier Score indicates accurate probability predictions.
- High ECE with high accuracy suggests *miscalibration* (e.g., underconfidence or overconfidence).

These metrics enable us to assess whether the model’s uncertainty estimates are meaningful and reliable, which is critical for clinical decision-support applications.

2.5 Method Overview

Figure 1 illustrates the overall architecture of the proposed system. The framework follows a retrieval-augmented multimodal pipeline that integrates visual features, textual knowledge, and

structured clinical information to generate accurate radiology reports.

The system performs **multi-signal hybrid retrieval** by combining image-based similarity, text-based similarity, and structured clinical labels (Problems). Retrieved items are treated as **case-level units** (image, report, and labels), ensuring clinical consistency. These multimodal features are integrated through an **attention-based fusion mechanism**, which dynamically learns the importance of different modalities and retrieved cases. The fused representation is then passed to the R2Gen generator to produce the final radiology report.

2.6 Implementation Detail

The proposed **Hybrid Retrieval-Augmented Generation (Hybrid RAG)** framework extends a standard image-to-text pipeline by integrating multimodal retrieval and attention-based fusion to improve report quality and clinical reliability.

Visual Encoding and Feature Alignment The BioViL-T encoder extracts visual embeddings from input chest X-ray images, producing feature representations:

$$V \in R^{N \times d}$$

Since these embeddings differ from the feature space expected by R2Gen, a linear projection layer is applied to map them into the latent space of the R2Gen model:

$$V' = WV$$

where W is a learnable transformation matrix ensuring dimensional compatibility.

Hybrid Retrieval To incorporate external knowledge, the model retrieves top-3 similar cases based on:

- visual similarity (image embeddings),
- semantic similarity (report embeddings),
- structured clinical labels (multi-hot vectors).

Each retrieved case consists of image features, report text, and clinical labels, preserving cross-modal consistency.

Multimodal Fusion and Generation The projected visual features (V'), retrieved representations (R), and label embeddings (L) are combined using an attention-based fusion mechanism:

$$F = \text{Attention}(V', R, L)$$

The fused representation F serves as the encoder output for the R2Gen decoder, which generates the radiology report in a sequence-to-sequence manner.

Training Configuration The model is trained using cross-entropy loss and optimized with the Adam optimizer. To ensure stable and effective training, we apply teacher forcing, dropout, gradient clipping, and early stopping based on validation performance.

This implementation enables effective integration of multimodal inputs and retrieved knowledge, leading to more accurate, coherent, and clinically grounded report generation.

2.7 Training Strategy

The proposed model is trained in a supervised manner using paired chest X-ray images and corresponding radiology reports. The **Findings** section serves as the ground truth target for report generation.

Dataset Split The preprocessed dataset of 3,337 samples is divided into three subsets:

- **Training set (70%)**: Used to learn model parameters.
- **Validation set (10%)**: Used for hyperparameter tuning and model selection.
- **Test set (20%)**: Used for final performance evaluation.

The split is performed at the patient level to avoid data leakage between sets.

Training Procedure Each sample includes frontal and lateral X-ray images, structured labels, and the target *Findings* report. Images are encoded using **BioViL-T**, and top-3 similar cases are retrieved based on image, text, and label similarity. Retrieved and input features are fused via attention and passed to the **R2Gen** decoder for sequence generation using teacher forcing.

Loss and Optimization The model is trained using **cross-entropy loss** and optimized with the **Adam optimizer**. Early stopping based on validation performance is applied.

Validation and Regularization Model performance is monitored using validation loss and BLEU scores. **Dropout**, **gradient clipping**, and **teacher forcing** are used to improve stability and generalization.

3 Results and Analysis

3.1 Comparison with Baseline Models

Table 1 presents a comprehensive comparison between the proposed model and established baseline approaches across lexical, clinical, and uncertainty-aware evaluation metrics.

From a lexical performance perspective, METransformer achieves the strongest results, particularly in BLEU-1 (0.483), BLEU-3 (0.228), and BLEU-4 (0.172), indicating superior fluency and n-gram consistency. R2Gen also performs competitively, especially in BLEU-2 (0.479). In contrast, the proposed model achieves moderate lexical scores (e.g., BLEU-4: 0.110, ROUGE-L: 0.315), suggesting that retrieval augmentation may slightly reduce exact textual overlap with reference reports.

However, lexical metrics alone do not reflect clinical usefulness. When considering clinical evaluation, METransformer achieves the highest RG-F1 score (0.311), demonstrating strong capability in capturing medical entities and relationships. R2GenCMN performs best in CheXbert-based metrics, achieving the highest 14Ma-F1 (0.097) and 14Mi-F1 (0.717), indicating strong disease classification performance, particularly for frequent conditions.

The proposed model demonstrates balanced performance across clinical metrics, with RG-F1 (0.235) and reasonable CheXbert scores (14Mi-F1: 0.356). While it does not outperform all baselines in absolute terms, it maintains competitive clinical grounding while integrating additional retrieval-based evidence.

A key contribution of this work is the inclusion of uncertainty and calibration metrics. In this regard, R2GenCMN achieves the best calibration performance (ECE: 0.040, Brier Score: 0.040), indicating highly reliable confidence estimates. The proposed model achieves moderate calibration performance (ECE: 0.076, Brier: 0.077), comparable to PPKED and METransformer. This suggests that while the model produces reasonably calibrated predictions, there is still room for improvement in uncertainty estimation.

Overall, the results highlight an important trade-off. Transformer-based models (e.g., METransformer) excel in lexical fluency and structured generation, while memory and classification-focused models (e.g., R2GenCMN) perform well in disease prediction and calibration. The proposed Hybrid RAG model achieves a balanced trade-off between

clinical grounding, multimodal reasoning, and uncertainty awareness, though it does not consistently achieve the highest score across all individual metrics.

3.2 Ablation Study

The ablation study (Table 2) evaluates the contribution of different components of the proposed framework.

The baseline configuration (A0: R2Gen) achieves strong lexical performance (BLEU-1/2: 0.479) and solid clinical scores, confirming its effectiveness as a foundational image-to-text model.

The A1 configuration, which relies primarily on structured clinical labels, shows a significant drop in lexical performance (BLEU-4: 0.041, ROUGE-L: 0.058). This demonstrates that structured labels alone are insufficient for generating coherent and contextually rich radiology reports. Interestingly, its 14Ma-F1 remains relatively stable (0.078), suggesting that label-based inputs may still capture some disease-level signals.

The A2 configuration introduces partial multimodal integration and shows clear improvement over A1 across all metrics, particularly in lexical (BLEU-4: 0.068) and clinical scores (RG-F1: 0.225), confirming the benefit of combining modalities.

The proposed full model achieves the best overall balance in the ablation setting, improving both lexical (BLEU-4: 0.110) and clinical performance (RG-F1: 0.254). This demonstrates that hybrid retrieval combined with multimodal fusion is essential for achieving robust performance.

Additionally, the inclusion of ECE and Brier Score provides further insight. A2 achieves the best calibration (ECE: 0.059, Brier: 0.058), while the proposed model shows slightly higher values (ECE: 0.076), indicating a trade-off between improved generation quality and calibration.

3.3 Key Observations

Several important patterns emerge from the results. First, there is a consistent trade-off between lexical similarity and clinical correctness, with models optimized for fluency not always achieving the best clinical grounding. Second, retrieval-based augmentation improves contextual relevance but may introduce variability in wording, which negatively impacts BLEU and ROUGE scores. Third, multimodal fusion significantly enhances performance compared to single-modality inputs, confirming

Model	BL-1	BL-2	BL-3	BL-4	ROUGE-L	RG-F1	14Ma-F1	14Mi-F1	ECE	Brier Score
R2Gen	0.479	0.479	0.103	0.082	0.483	0.242	0.079	0.376	0.088	0.087
R2GenCMN	0.362	0.267	0.203	0.153	0.545	0.254	0.097	0.717	0.040	0.040
PPKED	0.099	0.052	0.032	0.021	0.178	0.247	0.040	0.423	0.076	0.076
METransformer	0.483	0.322	0.228	0.172	0.480	0.311	0.074	0.080	0.078	0.079
Proposed Model	0.384	0.231	0.155	0.110	0.315	0.235	0.064	0.356	0.076	0.077

Table 1: Comparison of baseline models and the proposed Hybrid RAG framework. Best results are highlighted in bold.

Model	BL-1	BL-2	BL-3	BL-4	ROUGE-L	RG-F1	14Ma-F1	14Mi-F1	ECE	Brier Score
A0	0.479	0.479	0.103	0.082	0.483	0.242	0.079	0.376	0.088	0.087
A1	0.060	0.015	0.004	0.041	0.058	0.011	0.078	0.376	0.087	0.088
A2	0.221	0.146	0.102	0.068	0.302	0.225	0.053	0.250	0.059	0.058
Proposed Model	0.384	0.231	0.155	0.110	0.315	0.254	0.064	0.356	0.076	0.077

Table 2: Ablation study evaluating the contribution of different components. Best results are highlighted in bold.

the importance of integrating complementary information sources. Finally, calibration metrics reveal that better generation quality does not always correspond to better uncertainty estimation, highlighting an important research gap that warrants further investigation.

4 Discussion

The experimental results highlight several key insights into the performance of the proposed Hybrid Retrieval-Augmented Generation (Hybrid RAG) framework. Compared to baseline models, the proposed approach achieves competitive performance in clinical metrics, particularly RG-F1, indicating improved clinical grounding through the use of retrieved case-based evidence. However, the model shows relatively lower BLEU and ROUGE scores, suggesting that retrieval introduces variability in phrasing, which reduces lexical similarity with reference reports.

The ablation study demonstrates the importance of multimodal fusion. Models relying only on structured labels perform poorly, while the integration of visual features, labels, and retrieved cases leads to consistent improvements in both lexical and clinical performance. This confirms that combining multiple modalities provides complementary information necessary for generating coherent and clinically meaningful reports.

In terms of reliability, the proposed model achieves moderate calibration performance based on ECE and Brier Score. While comparable to several baselines, it does not outperform the best-calibrated model, indicating that uncertainty estimation remains a challenge. Additionally, the results show that improved generation quality does not always correspond to better calibration.

Despite these contributions, several limitations exist. Lexical metrics may not fully reflect clinical correctness, the retrieval mechanism may introduce irrelevant information, and calibration performance is not yet optimal for clinical deployment. Furthermore, the use of a single dataset may limit generalizability. Overall, the findings demonstrate the effectiveness of retrieval and multimodal integration while highlighting important areas for future improvement.

5 Conclusion

This work presents a Hybrid Retrieval-Augmented Generation (Hybrid RAG) framework for radiology report generation, integrating visual features, structured clinical labels, and retrieved case-based knowledge. The proposed approach improves clinical relevance and maintains competitive performance across evaluation metrics, particularly in capturing meaningful medical information.

The ablation study confirms the importance of multimodal fusion and retrieval in enhancing report quality. While the model achieves reasonable calibration performance, uncertainty estimation remains an area for improvement.

Overall, the results demonstrate the potential of combining retrieval and multimodal learning for more reliable radiology report generation, with future work focusing on improving retrieval quality, model calibration, and generalization to broader clinical settings.

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